

MACHINE LEARNING-BASED STUDENT PERFORMANCE AND DROPOUT RISK PREDICTION FOR EARLY IDENTIFICATION OF AT-RISK STUDENTS

PHASE I REPORT

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BONAFIDE CERTIFICATE

Certified that this Report titled “**MACHINE LEARNING-BASED STUDENT PERFORMANCE AND DROPOUT RISK PREDICTION FOR EARLY IDENTIFICATION OF AT-RISK STUDENTS**” is the bonafide work of **Musa Haji (230701391), Prithivi Raj (230701508) and Sachin Chaudhary (230701402)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

“This project addresses the following Sustainable Development Goal(s):

Goal No: 4 – Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Goal No: 10 – Reduce inequality within and among countries by identifying students who are at academic risk and supporting timely interventions to improve educational outcomes.”

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ABSTRACT

With the growing access to student academic and behavior data, it is now possible to quickly identify which students are likely to face academic challenges. Historically, most approaches to assessing student performance involve manual grading and tracking attendance for a deep analysis of performance metrics. However, this approach does little to help uncover complex correlations between various academic, demographic, and behavioral characteristics of students. To solve this problem caused by traditional assessment methods, this research project offers to develop a machine learning system for early identification of at-risk students called Machine Learning-Based System for predicting student performance and dropouts. The system will analyze students' academic, personal, and behavioral information to categorize their performance level. The system's operational process starts with data preparation processes including, among others, filling the blanks in the data set, excluding the contradictory points, transforming categorical information, and so on. A variety of classification techniques will be covered to accomplish the task of predicting students' performance, which includes logistic regression, decision trees, random forests, k-nearest neighbors, support vector machines, and gradient boosting. Stratified cross-validation and hyperparameter optimization have been used for the purposes of having as reliable a model as possible and recognizing the most suitable one among those tested. Based on the results obtained as a result of conducting experiments, Gradient Boosting turned out the most suitable model with 87.34% accuracy, 87.95% precision, 87.34% recall, and 87.42% F1-score. The feature importance analysis found that prior performance in school contributes significantly to the prediction made by the system and the G2 grade is the most important criterion, while the G1 grade, age, behavior at school, drinking habits, parents' profession, and failures also affect the outcome. Even though the present dataset does not have a particular dropout-marked objective, the created predictive framework provides a basis for the advanced dropout-risk prediction system. The coming development will involve the application of designated dropout data which will make it possible to provide accurate dropout-risk prediction.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
API	Application Programming Interface
AUC	Area Under the Curve
CGPA	Cumulative Grade Point Average
CSV	Comma-Separated Values
CV	Cross-Validation
DFD	Data Flow Diagram
DT	Decision Tree
EDM	Educational Data Mining
GB	Gradient Boosting
KNN	K-Nearest Neighbors
LR	Logistic Regression
ML	Machine Learning
RF	Random Forest
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
UCI	University of California, Irvine
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The ever-increasing access to educational data has brought with it various possible ways of examining student performance. Educational institutions are constantly gathering information regarding their students' records, attendance, demography, family background, social life, and other behavioral aspects. Manual data processing can be tiresome and rather ineffective in revealing trends that help to determine a student's performance in the future.

Student performance is influenced by numerous factors, thus it is essential to identify students in need of assistance as soon as possible for effective help. Traditional performance evaluation methods rely mainly on test scores and results of periodical tests, which may not be enough to recognize the hidden patterns in the students' performance. The use of machine learning algorithms is necessary for successful examination of historical student records. The framework developed establishes a basis for detecting students who may need additional academic assistance early on. Also, an appropriate dropout-tagged dataset could be added to the framework in order to make predictions about dropout probability and assist educational institutions in implementing progress strategies.

1.2 OBJECTIVE

The primary objective of the system being proposed is to implement a machine learning-enabled system for predicting students' performance and possible dropout risks, which may aid in understanding and analyzing all student-related academic, demographic, social and behavioral aspects.

The specific objectives are:

1. To gather student academic, social, demographic, and behavioral data for analysis.
2. To process the collected student data to deal with missing values, inconsistent records and duplication of data.
3. To encode categorical attributes and convert numerical data into a form that can be used in machine learning.
4. To create training and testing datasets out of the processed data so it can be utilized in model development and evaluation.
5. To determine the factors that may affect the academic performance of students.
6. To implement machine learning classification algorithms.
7. To assess the model performance through Accuracy, Precision, Recall, and F1-score evaluation.

8. To perform stratified cross-validation to test how reliable and consistent the produced models are.
9. To execute hyperparameter tuning for making the machine learning models more effective.
10. To categorise students into the Low, Average, and Good performance groups.
11. To calculate the prediction probability for every individual student.
12. To employ feature analysis to identify the factors that influence the predictions of student performance.
13. To provide an early identification mechanism for students who may require additional academic guidance and intervention.
14. To establish a foundation for extending the system with a dedicated dropout-risk prediction module using an appropriately labelled dropout dataset.

1.2.1 IMPLEMENTATION OBJECTIVE

For the present project implementation, the scope is restricted to:

- Student Data Collection and Preprocessing
- Categorical Encoding and Numerical Feature Transformation
- Student Performance Classification
- Machine Learning Model Training and Comparison
- Cross-Validation and Hyperparameter Tuning
- Performance Evaluation using Accuracy, Precision, Recall, and F1-score
- Feature-Importance Analysis
- Individual Student Performance Prediction
- Prediction Probability Generation
- Early Identification of Students Requiring Academic Support

1.3 EXISTING SYSTEM

Multiple approaches and machine learning models have been developed for analysing and predicting student academic performance. Traditional educational systems generally depend on examination marks, attendance records, assignments and periodic assessments to evaluate students. Statistical techniques and conventional classification algorithms are also used to analyse student-related data and identify factors that influence academic outcomes. Datasets such as the Student Performance dataset provide academic, demographic, social and behavioural attributes that can be used for predicting student performance.

Apart from traditional statistical approaches, various machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine and Gradient Boosting have been applied to student performance prediction. These models can identify patterns within historical student records and classify students according to their expected academic performance. However, the effectiveness of these approaches depends heavily on the quality of the dataset, appropriate preprocessing, feature selection and the distribution of the target classes. In cases where some performance categories contain very few students, class imbalance can negatively affect the prediction results.

The major limitation of existing student performance assessment systems is that they primarily focus on analysing academic results after the completion of assessments rather than providing an early indication of students who may require support. Manual analysis of multiple academic, demographic and behavioural factors can also be time-consuming and may overlook relationships between different attributes. Furthermore, many prediction systems provide only the final classification without clearly indicating which factors contributed to the prediction. Therefore, a system that combines machine learning-based prediction with model evaluation and feature-importance analysis is highly useful for supporting early identification of students who may need academic intervention.

1.4 PROPOSED SYSTEM

The proposed system is a machine learning-based student performance and dropout risk prediction framework. The system accepts student academic, demographic, attendance, behavioural, social, and family-related data and performs the high-level operations shown in Figure 1.4.

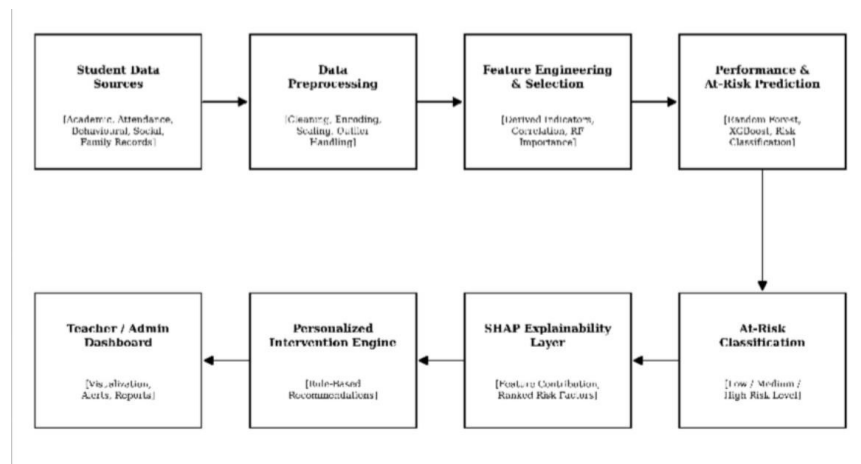


Figure 1.4: System Architecture

The system consists of the following major conceptual components:

1. Student Data Collection

2. Data Preprocessing
3. Feature Engineering and Selection
4. Performance Prediction
5. Risk Classification
6. Explainability Layer
7. Intervention Support
8. Teacher/Administrator Dashboard

The current implementation concentrates on Data Preprocessing, Performance Prediction, Model Evaluation, and Feature Importance Analysis.

Data Preprocessing

The module:

- cleans student data,
- handles missing values,
- encodes categorical features,
- prepares numerical features,
- divides the dataset for training and testing.

Performance Prediction

The prediction module:

- trains different machine learning models,
- compares their performance,
- performs cross-validation and tuning,
- predicts student performance as Low, Average, or Good.

Explainability

The explainability layer identifies the important factors contributing to a student's prediction and helps provide an understandable basis for the model's output.

Risk Identification

The system uses student performance and relevant factors to identify students who may require additional academic attention. The framework can later be extended to direct Low / Medium / High dropout-risk prediction using a suitable dropout-labelled dataset.

CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE REVIEW

The growing popularity of educational data and machine learning approaches has led to a substantial amount of interest in predicting the academic performance of students and identifying those who need academic support from an early age. Prior studies have utilized a variety of factors including academic, demographic, behavioral, and social for predicting student outcomes. This literature review highlights the important publications regarding student performance prediction, classification using machine learning, features, and early at-risk identification.

2.1.1 STUDENT PERFORMANCE DATASETS AND ACADEMIC PREDICTION

Performance prediction of students was done by Cortez and Silva (2008) based on the information obtained from secondary schools. The study focused on the application of machine learning approaches to make predictions about the grades of students and also identified factors linked with academic performance of students.

The UCI Student Performance dataset is another dataset which has been extensively used for the purpose of making predictions about the performance of students. The dataset contains information about the academic performance of students, their family backgrounds, social activities, study habits and many other behavioral factors.

Machine learning methods that have been used for student performance classification include Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Gradient Boosting. These help to recognize the patterns in the student data and classify the students based on their potential academic performance.

Machine learning techniques like Random Forest and Gradient Boosting are more suitable as they take into account the information provided by several different features as well as the non-linear relationships in the dataset. However, the prediction accuracy is influenced by class imbalance, quality of features and suitable model parameters.

2.1.2 FEATURE IMPORTANCE AND EXPLAINABLE PREDICTION

Academic predictive models must not only give a prediction result but also help educators to better understand what affects the given prediction result. Feature-importance methods can be used to determine which attributes affect the model the most. Explainable AI approaches like SHAP (SHapley Additive exPlanations) allow interpreting a prediction result in terms of contributions made by particular features. Such methods may enhance the

transparency of machine learning-driven education-related models and help to get more useful predictions from them.

Identifying students who might suffer poor academic results early on is one of the key applications of educational data mining. Based on the analysis of past academic performance and student attributes, prediction methods can help detect students who will need extra assistance.

However, most of the existing methodologies concentrate mostly on predicting grades or overall academic performance rather than developing an early warning system that includes all the steps necessary. Moreover, predicting dropouts directly implies the availability of a dataset with labelled dropout records.

2.2 COMPARISON AND DISCUSSION

Study / System	Performance ML Prediction	Feature Analysis	Early Identification	Explainability
Cortez & Silva – UCI Student Performance Dataset	✓	Limited	Limited	✗
Traditional ML Approaches	✓	Limited	✗	✗
SHAP-based Approaches	✓	✓	Limited	✓
Proposed System	✓	✓	✓	✓

Table 2.2: Comparison of Existing Systems

2.3 CONCLUSION

Literature shows that machine learning tools could be efficiently applied for student data analysis and academic performance prediction. Current research indicates the significance of academic, demographic, social, and behavioral features of students and the influence of these features on the performance of students. Different classification models have also been used for prediction of student performance in current research, which makes a good basis for the development of predictive educational systems.

However, current systems pay much attention to prediction and offer little information on the factors that determine the predicted result. In addition, there is a need for systems that will be able to find students in need of academic assistance before examinations, not after their results. Explainable machine learning methods will add additional value to such systems through identification of factors influencing the prediction.

The comparison and evaluation of models are essential steps for the selection of a reliable prediction mechanism. Various methods such as cross-validation and hyperparameter optimization will facilitate the estimation of consistency in the models and

the improvement of the predictions' performance. Moreover, the extraction of important features will facilitate the understanding of factors that influence student results more.

In conclusion, the proposed solution combines data pre-processing, prediction of student performance through machine learning techniques, evaluation of the model, feature importance estimation, and detection of students in need of academic assistance. The solution serves as a basis for data-driven educational interventions and can contribute to making decisions about student assistance. In addition, the proposed solution can be expanded by predicting the risk of student dropout through an appropriate labelled dataset.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

Figure 3.1 presents the architecture of the Machine Learning-Based Student Performance and Dropout Risk Prediction System. The process starts from the data source that consists of data about academics, attendance, behaviour, social and family details of students. The Data Preprocessing Module cleans and scales the data along with feature engineering and selection for determining the right features.

The selected features are fed into the Performance and At-Risk Prediction module which employs machine learning models like Random Forest and XGBoost. The next stage is At-Risk Classification to determine the risk of the particular student. In the next layer the SHAP Explainability Layer determines the important predictors and in the Personalized Intervention Engine, the right suggestions are provided. The Teacher/Administrator Dashboard provides visual representations to the users.

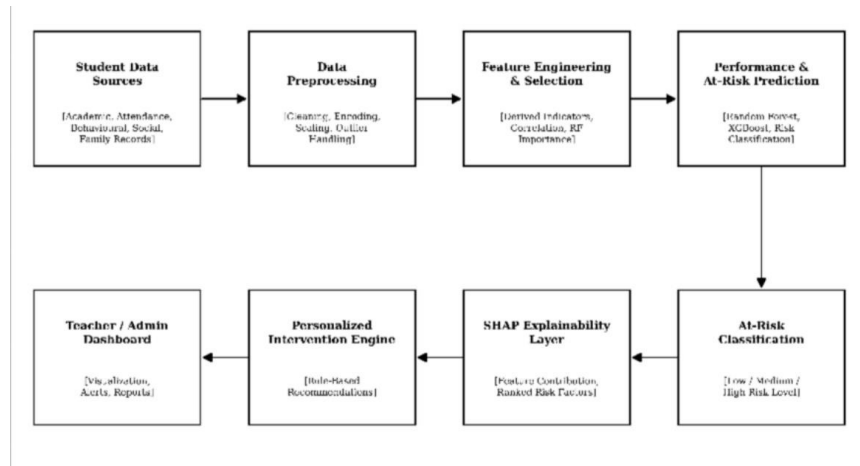


Figure 3.1: Overall architecture of the proposed machine learning-based student performance and dropout risk prediction system

The first component of the architecture is the Student Data Sources component. This component serves as an input layer of the architecture and supplies the necessary information for predictions. Such information may comprise academic data, attendance data, behavioral attributes, social attributes, and family attributes. Different attributes enable the architecture to take into consideration several aspects of a student's profile and academic performance instead of relying solely on the exam scores.

The collected information is supplied further to the Data Preprocessing component. The raw data about the students may have missing values, categorical attributes, inconsistencies, and attributes with different scales. In order to apply machine learning methods to the data, preprocessing is carried out first. The processes involved include data cleaning, categorical attribute transformation, scaling, and other necessary data processing.

Next, the data goes to the Feature Engineering and Selection module. At this step, the goal is the discovery of features that will help predict student performance. The derived indicators can be calculated based on available features, and the contribution of each can be studied using correlation analysis or by evaluating feature importance from the models. Feature selection allows the prediction step to consider only valuable information about students and minimize the impact of irrelevant information.

The resulting features are sent to the Performance and At-Risk Prediction module. This is the machine learning step of the architecture. Models are built using the processed dataset of students, and learn the relationship between inputs and outputs. Models like Random Forest or XGBoost can be employed for prediction at this step. At the end, the result of the prediction stage is the performance output as well as additional information needed to find at-risk students.

The output of the prediction goes through the At-Risk Classification module. This is a phase where the output of the model is transformed into risk categories. The architecture employs risk levels known as Low, Medium, and High Risk Level. It makes the prediction more interpretable and enables the educators to prioritize the students who need additional attention.

This leads to the SHAP Explainability Layer. The fact that a prediction was made does not by itself show why a certain student received it. The explanation comes from the analysis of the contribution of each individual feature to the model prediction. SHAP-based feature contribution helps in identifying and ranking the features which contribute greatly to the prediction for a certain student.

Information that is generated in the prediction and explainability processes is provided to the Personalized Intervention Engine. This engine utilizes the risk factors and predictions generated from the previous stages to formulate rules, for instance to identify students in need of academic intervention. The objective of this process is to derive intervention information from the results of predictions and not only classify the problem.

The last process involves the presentation of the results in the Teacher/Administrator Dashboard. The dashboard is the output and monitoring part of the framework and provides information on the performance and risk of students along with visualizations.

In summary, the architectural design provides for a full flow of student data acquisition → data preprocessing → feature extraction → prediction through machine learning → risk categorization → explanations → interventions → dashboard visualizations. The current design mainly focuses on the establishment of the data processing and student performance prediction process, but the architecture provides the basis for expanding the process to predict dropouts using appropriately labelled data.

3.2 SYSTEM REQUIREMENTS

This section covers the Software Requirements, Hardware Requirements and Dataset Requirements needed to build the proposed Machine Learning-Based Student Performance and Dropout Risk Prediction System.

3.2.1 SOFTWARE REQUIREMENTS

Component	Requirement
Operating System	Windows / Linux
Programming Language	Python
Development Environment	Jupyter Notebook / Google Colab
ML Framework	Scikit-learn
Visualization	Matplotlib, Seaborn
Explainable AI	SHAP
Data Storage	CSV / Pandas DataFrame
Input Format	CSV
Output Format	Prediction Report / DataFrame
Prediction Output	Low / Average / Good
Prediction Probability	Class Probability Scores

Table 3.2.1: Software Requirements

SHAP is a part of the explainability module because it allows us to determine the role of each particular feature of the students in the outcome of the prediction by the model. We will be able to see the role played by such characteristics as grade, age, absences, and failures in the prediction of academic performance.

3.2.2 HARDWARE REQUIREMENTS

Provisioning of the hardware resources is necessary in order to ensure that the development, testing, and application of the student performance predictor works smoothly. The machine learning pipeline includes data preprocessing, feature transformation, model training, cross-validation, hyperparameter tuning, and prediction, which all require sufficient CPU and memory. As the models were constructed using traditional machine learning methods, there is no need for a dedicated GPU. Table 3.2.2 shows the hardware configuration that is necessary to run the system. The minimum configuration is sufficient for development, data preprocessing, model training, debugging, and testing, whereas the recommended one allows additional computation and memory capacity for further model training, cross-validation, and larger datasets.

Component	Minimum Configuration	Recommended Configuration
Processor (CPU)	Intel Core i3 / Equivalent Multi-Core CPU	Intel Core i5 or above
System Memory (RAM)	4 GB	8 GB or above
Graphics (GPU)	Not Required	Optional NVIDIA GPU
Storage (SSD)	256 GB SSD	512 GB or above SSD

Table 3.2.2: Hardware Requirements

3.2.3 DATASET REQUIREMENTS

The Student Performance Dataset includes:

- Academic and demographic information of students
- 395 student records
- 33 original features
- 32 input features after target removal
- 58 processed features after encoding and transformation
- 316 training records
- 79 testing records
- Low, Average, Good – performance classes

The Student Performance Dataset is used to design and test the machine learning-based student performance prediction system. It consists of academic, demographic, social, and behavioral characteristics of students which help in identifying the factors that affect academic performance. During preprocessing, the missing and inconsistent values are handled, the categorical attributes are encoded, and the input attributes are prepared in the desired format for machine learning modeling.

The four classes in the original target attribute, i.e., Low, Average, Good, and Excellent, have been reduced to three classes because only 19 records belong to the Excellent category, which creates severe class imbalance. The three target classes are now Low, Average, and Good. The models are tested on accuracy, precision, recall, and F1-score. The current dataset is used for predicting student performance; however, a separate dropout-labelled dataset is needed for predicting student dropouts.

3.3 ACTIVITY DIAGRAM

The activity diagram in Figure 3.3 shows the workflow of the proposed system. The process starts by loading the student dataset, which is then preprocessed (cleaning, encoding and scaling) and passed through feature selection and engineering. Two parallel activities follow: the machine learning models (Logistic Regression, Decision Tree, Random Forest, KNN, SVM and Gradient Boosting) are trained, and cross-validation with

hyperparameter tuning is carried out. The models are then evaluated using Accuracy, Precision, Recall and F1-score, predictions are generated as Low, Average or Good, the results are explained using SHAP feature importance, and finally the output is stored and displayed as a dashboard or report.

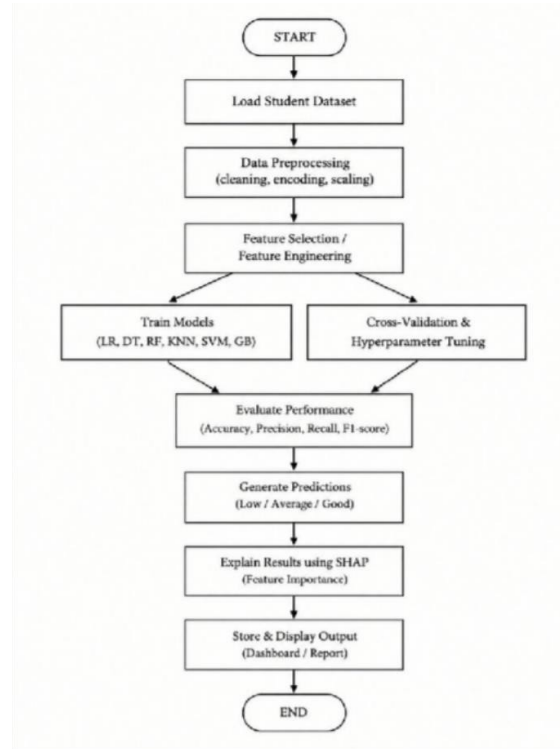


Figure 3.3: Activity Diagram

3.4 DATA FLOW DIAGRAM

The Data Flow Diagram depicts the overall process flow of the student performance prediction system from the student dataset and data collection to data processing, after which data goes through feature selection and training of the model. At the same time, cross-validation and hyperparameter optimization are done to optimize and assess the models. Feature analysis using SHAP is applied to the processed features to detect the most impactful factors on the predictions. As a result, the model predicts the student's performance as Low, Average, or Good with probabilities and gives the final result.

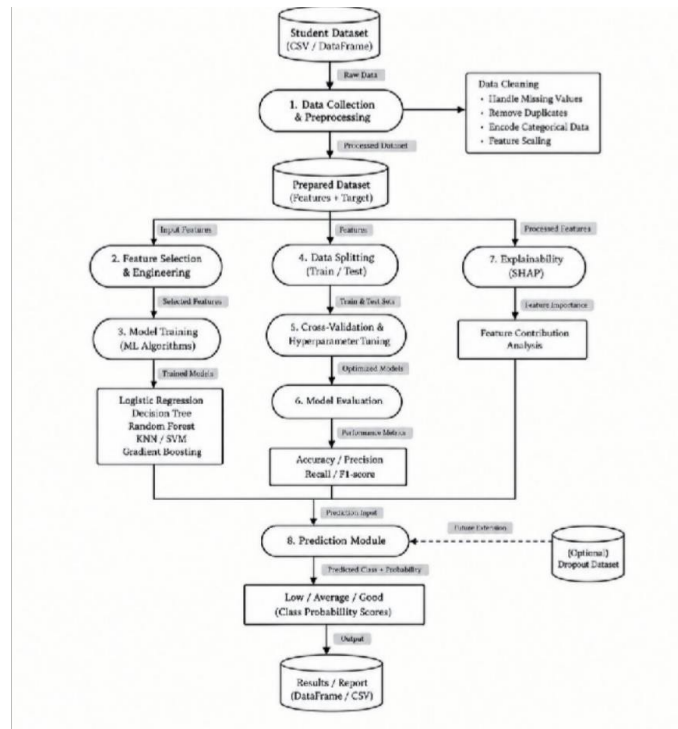


Figure 3.4: Data Flow Diagram

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

The suggested methodology is based on a modular prediction approach that utilizes machine learning algorithms. Instead of employing a single classification method for predicting students' performance, it breaks the whole process down into several steps, comprising data preprocessing, feature generation, training of the prediction model, evaluation of the model, prediction, and explanation.

4.1.1 MACHINE LEARNING-BASED PERFORMANCE PREDICTION

The prediction module performs the entire process of training and evaluating machine learning models from the preparation of student data to the generation of the final performance category. Different classification algorithms are utilized and assessed to find the best model for predicting the academic results of students. Thus, this provides a structured evaluation of various models and allows for reliable identification of students with academic difficulties.

4.1.2 DATA PREPROCESSING AND FEATURE ENGINEERING

The processing of student information takes place prior to its consumption by the prediction models. The data processing stage deals with the absence of data, eliminates duplicate and/or inconsistent information, transforms the values, and applies the appropriate rules for encoding categorical variables. The processing proceeds in the following order:

Stage 1: Data Cleaning

The preliminary student dataset is cleansed by dealing with missing data, removing duplicates, and making the data consistent. This makes sure that the data fed into the machine learning model is appropriate for analysis.

Stage 2: Feature Transformation

Categorical features are transformed into numbers, and then these numbers undergo scaling. After that, feature generation and selection are carried out to prepare a relevant feature set for building the model.

Stage 3: Model Training and Evaluation

The processed dataset is split into a training set and a test set. The models used are Decision Trees, Logistic Regression, Random Forests, KNN, SVMs, and Gradient Boosting, which are trained and evaluated on the basis of their accuracy, precision, recall, and F1-score. Cross-validation and hyperparameter tuning were used to improve the performance of the models.

Stage 4: Explainable Prediction

SHAP is utilized in understanding the influence of individual features on the output. This analysis allows one to detect the relevant parameters impacting students' performance. The result of the system denotes whether the predicted category is Low, Average, or Good and shows the respective probabilities.

4.2 MODULES

4.2.1 DATA COLLECTION AND PREPROCESSING MODULE

The Data Collection and Preprocessing Module is responsible for receiving the student performance dataset and preparing it for machine learning.

Functions

- Dataset loading
- Missing value handling
- Duplicate removal
- Inconsistent data handling
- Categorical data encoding
- Numerical feature transformation
- Feature scaling
- Train-test data preparation

The module takes unprocessed records of students as input which it processes into a formatted dataset that can be used for machine learning. The dataset comprises student characteristics related to academics, demography, sociology and behavior and goes through cleaning before proceeding to further processing.

Output

Processed Student Dataset → Cleaned Features + Target

4.2.2 FEATURE SELECTION AND ENGINEERING MODULE

This module identifies the most important predictors of student academic outcomes. Categorical attributes undergo transformation to be given their numerical forms after preprocessing.

A. Feature Transformation

Categorical variables are encoded into numerical values so that they can be processed by machine learning algorithms. Numerical attributes are also transformed and scaled where required.

B. Feature Selection

The module finds suitable student characteristics that affect academic performance. Specific aspects taken into account in the model include the past grades, age, attendance, previous failures and behavioural traits. The processed dataset has 58 features after the encoding and transformation operations are complete.

4.2.3 MACHINE LEARNING MODEL TRAINING

The Machine Learning Model Training Module is in charge of training a number of classification methodologies utilizing the prepared dataset of students. The models that are trained include:

- Logistic Regression
- Decision Trees
- Random Forests
- K-Nearest Neighbors
- Support Vector Machines
- Gradient Boosting

The dataset is split into 316 records used for training and 79 records meant for testing. All models are trained on the training data and later tested with the testing data. The target variable is classified into three performance categories: Low, Average and Good.

Initially, the four-class formulation featured an Excellent category, which encompassed just 19 students, causing a severe case of class imbalance. Hence, a three-class formulation was utilized.

4.2.4 CROSS-VALIDATION AND HYPERPARAMETER TUNING

The Cross-Validation and Hyperparameter Tuning Module enhances the efficiency and reliability of the learning models.

A. Stratified Cross-Validation

Performing five-fold stratified cross-validation allows the percentages of performance classes to be maintained across the folds. This approach makes it possible to obtain more reliable estimates than using only a single train-test split.

B. Hyperparameter Tuning

Grid Search is carried out for the selected machine learning algorithms to find the best parameters using cross-validation. The highest cross-validation F1-scores obtained were 0.8700 for Random Forest, 0.8803 for Decision Tree, and 0.8891 for Gradient Boosting.

4.2.5 MODEL EVALUATION MODULE

The Model Evaluation Module compares the performance of the trained classification models using standard evaluation metrics: Accuracy, Precision, Recall and F1-score.

The test dataset is used to evaluate the models and their capacity to classify student performance into categories. The final model selected is the tuned Gradient Boosting model, which has reached an Accuracy of 87.34%, Precision of 87.95%, Recall of 87.34%, and F1-score of 87.42%. This evaluation gives grounds to select the model which is best suited for the final prediction module.

4.2.6 STUDENT PERFORMANCE PREDICTION MODULE

Using the chosen machine learning model, the Student Performance Prediction Module makes predictions on newly entered student records. The input student features are processed in the same manner as they were prepared for model training, using the same preprocessing and feature transformation methods. After that, the trained Gradient Boosting model is utilized to predict in which academic performance category the student falls. The flow is: Student Input → Preprocessing → Trained Model → Performance Prediction → Low / Average / Good.

4.2.7 EXPLAINABLE AI MODULE

The Explainable AI Module serves to clarify which elements impact the machine learning model's outcomes. For that purpose, the SHAP (SHapley Additive exPlanations) method is used to analyze feature importance and make predictions more interpretable.

This module presents the features that most affect student performance. G2 is shown to be the leading factor in the model created, followed by G1, age, going-out frequency, alcohol consumption, parental occupation, absences and past failures. This module makes the prediction process more transparent as it elucidates the reasons behind a prediction instead of just providing the final classification.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION AND RESULTS

This chapter describes how the Machine Learning-Based Student Performance and Dropout Risk Prediction System has been implemented. Key features of the developed system include student data management, academic performance prediction, risk analysis, comparison of models, and explanation of predictions. The implementation provides an interactive interface which enables the administrator to access and evaluate information about students, conduct predictions and carry out model analysis of student performance.

5.1.1 DATA PRE-PROCESSING

The initial step in the pipeline is the conversion of the raw student dataset to a structured and clean format that is appropriate for machine learning. The system carries out the following preprocessing operations:

- **Handling Missing Values:** Identification and processing of missing values in order to keep consistency in the dataset.
- **Removal of Duplicates:** Elimination of duplicate and inconsistent records to avoid unnecessary bias in model training.
- **Encoding of Categorical Features:** Transformation of categorical attributes of students into a numerical form suitable for machine learning algorithms.
- **Transformation of Features:** Transformation of input features into a unified numerical format.
- **Scaling of Features:** Scaling of numerical features wherever needed so that they can serve as input data for the KNN, SVM, and Logistic Regression algorithms.

After preprocessing and encoding, the 33 original features transform into 58 processed features that are used for model training.

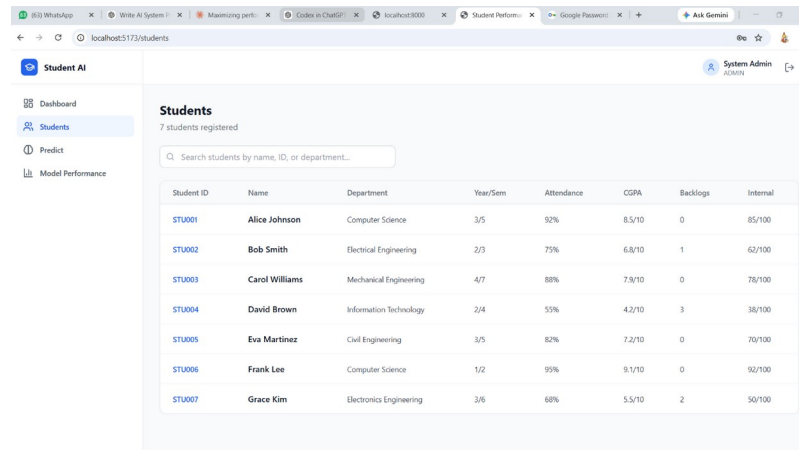
5.1.2 DATASET PREPARATION AND STUDENT MANAGEMENT INTERFACE

The Student Performance Dataset contains 395 student records with academic, demographic, social, and behavioural attributes. The processed dataset is divided into training and testing subsets for model development and evaluation.

Dataset Component	Records / Features
Total Records	395
Original Features	33
Processed Features	58
Training Records	316
Testing Records	79

Table 5.1.2: Dataset Preparation

The system also includes a Student Management Interface that lets the administrator browse and manage student records. In the interface shown in Figure 5.1.2, the administrator can view details such as Student ID, Name, Department, Year/Semester, Attendance, CGPA, Backlogs, and internal marks. A search tool has also been introduced that permits searching for students by name, ID, and department.



Student ID	Name	Department	Year/Sem	Attendance	CGPA	Backlogs	Internal
STU001	Alice Johnson	Computer Science	3/5	92%	8.5/10	0	85/100
STU002	Bob Smith	Electrical Engineering	2/3	75%	6.8/10	1	62/100
STU003	Carol Williams	Mechanical Engineering	4/7	88%	7.9/10	0	76/100
STU004	David Brown	Information Technology	2/4	55%	4.2/10	3	38/100
STU005	Eva Martinez	Civil Engineering	3/5	82%	7.2/10	0	70/100
STU006	Frank Lee	Computer Science	1/2	95%	9.1/10	0	92/100
STU007	Grace Kim	Electronics Engineering	3/6	68%	5.5/10	2	50/100

Figure 5.1.2: Student Management Interface

The student management component serves as the main source and supervision configuration for the system. It enables the categorization of student records and allows for prior review before making predictions. The information shown gives an organized form of academic parameters that can be processed further by the system.

5.1.3 MACHINE LEARNING MODEL RESULTS

The system assesses various machine learning classification algorithms to determine which one is the most appropriate for predicting student academic performance. The models utilized are Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine and Gradient Boosting.

Model	Accuracy	F1-Score
Decision Tree	87.34%	87.31%
Logistic Regression	86.08%	86.19%
Random Forest	86.08%	86.16%
Gradient Boosting	84.81%	84.89%
SVM	78.48%	78.36%
KNN	70.89%	71.07%

Table 5.1.3: Machine Learning Model Performance

The initial results show that the Decision Tree delivered the best test accuracy amongst the untuned algorithms, while the rest of the algorithms showed varying levels of performance across the different student performance classes.

5.1.4 CROSS-VALIDATION RESULTS

To assess the consistency of the models across different subsets of the dataset, 5-fold stratified cross-validation was performed.

Model	Mean F1-Score	Standard Deviation
Random Forest	0.8534	0.0653
Gradient Boosting	0.8509	0.0297
Decision Tree	0.8342	0.0237
Logistic Regression	0.8312	0.0486
SVM	0.7979	0.0306
KNN	0.6637	0.0394

Table 5.1.4: 5-Fold Stratified Cross-Validation Results

The cross-validation results indicate that Random Forest has the highest mean F1-score among the untuned models, and Gradient Boosting also performed well with less variability across folds.

5.1.5 HYPERPARAMETER TUNING AND FINAL MODEL EVALUATION

Grid Search was applied to selected machine learning models to identify suitable hyperparameter combinations and improve their predictive performance.

Model	Best Cross-Validation F1-Score
Random Forest	0.8700
Decision Tree	0.8803
Gradient Boosting	0.8891

Table 5.1.5(a): Hyperparameter Tuning Results

The optimized Gradient Boosting model reached the top cross-validation F1-score of 0.8891 and was chosen as the final academic performance prediction model. The chosen model was subsequently tested on the testing set.

Metric	Result
Accuracy	87.34%
Precision	87.95%
Recall	87.34%
F1-Score	87.42%

Table 5.1.5(b): Final Model Evaluation Results

The final model achieves an accuracy of 87.34% and an F1-score of 87.42%, indicating that it can classify students into the performance categories reliably.

5.1.6 CLASSIFICATION DASHBOARD AND RISK ASSESSMENT ANALYSIS

The developed system has an administrative dashboard, which provides consolidated information about students' performance and other prediction-related information.

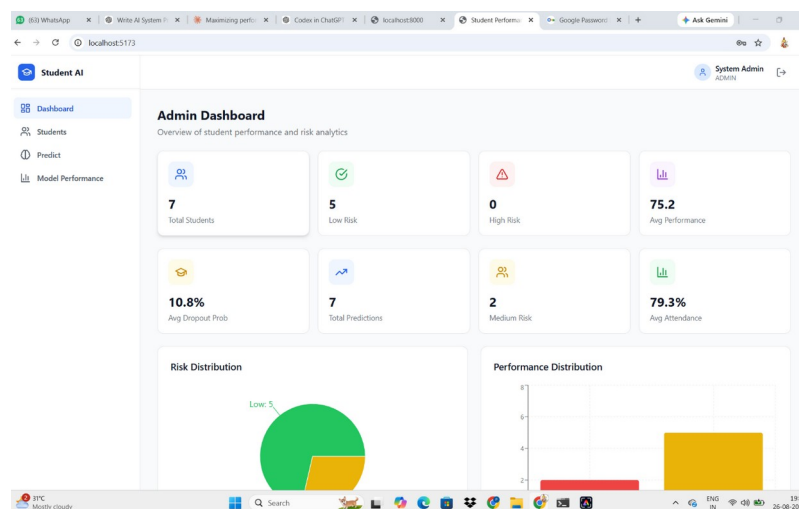


Figure 5.1.6: Administrative Dashboard and Risk Analysis

The dashboard shows key indicators at the system level, such as the total number of registered students, the number of students categorized as low-risk, medium-risk, and high-risk respectively, the average academic performance and the average dropout probability. In the example shown, the dashboard has 7 students in total, of which 5 belong to the Low Risk category and 2 belong to the Medium Risk category; no student is in the High Risk category. The visual analytics area has two charts, Risk Distribution and Performance Distribution, which summarize the outcomes of predictions graphically.

5.1.7 RECENT PREDICTIONS AND PREDICTION HISTORY

The prediction history interface provides information about previous prediction results.

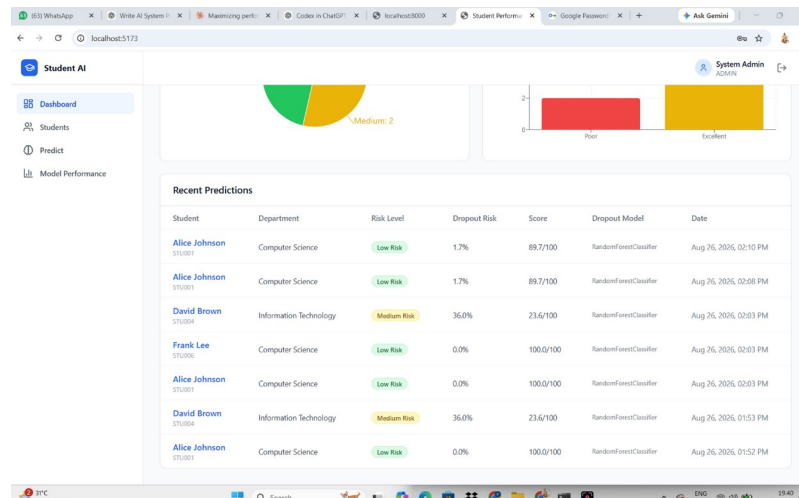


Figure 5.1.7: Recent Student Predictions

The information is displayed in a tabular format and has the following fields: student name, ID, department, risk level, dropout probability, academic performance score, model, and date and time of the prediction. The interface allows individual predictions to be tracked after the model has been executed. For instance, it labels students as “Low risk” or “Medium risk” and provides the corresponding numerical values of dropout risk. This feature keeps a history of predictions and allows users to compare results of students over different time frames.

It should be noted that the current dataset does not have a labelled dropout target. This interface is therefore a risk monitoring and output component of the implementation, and the dropout percentages shown are not the results of a validated dropout model.

5.1.8 MODEL PERFORMANCE COMPARISON

The implementation includes a dedicated Model Performance interface for comparing the machine learning algorithms used within the system.

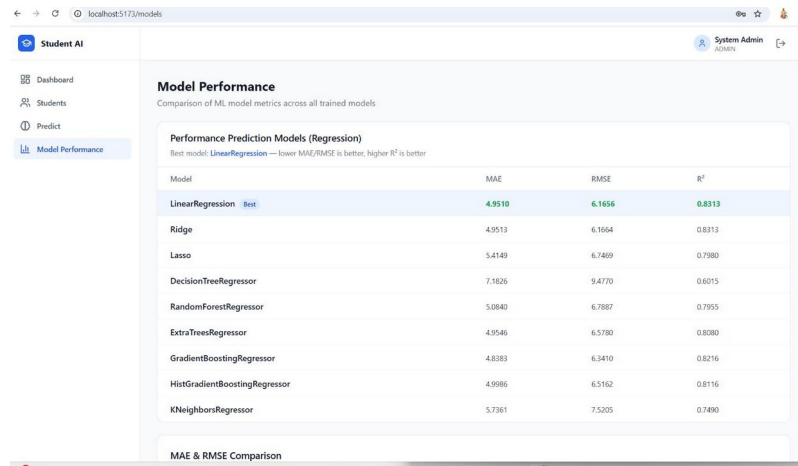


Figure 5.1.8: Machine Learning Model Performance Comparison

The interface shows the performance of many classification models with the help of tables and graphs. The models that can be compared include algorithms such as Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, AdaBoost, K-Nearest Neighbors, Support Vector Classifier, and Histogram Gradient Boosting, according to the comparison configuration selected by the user.

The graphical display enables the administrator to see the differences between evaluation metrics such as Accuracy, F1-score, and ROC AUC, and is more convenient than comparing numerical values alone. The model comparison module supports the choice of the best algorithm based on multiple evaluation metrics rather than accuracy only.

5.1.9 PREDICTION EXPLAINABILITY USING SHAP

To improve the transparency of the prediction process, the system incorporates a SHAP-based Prediction Explanation module.

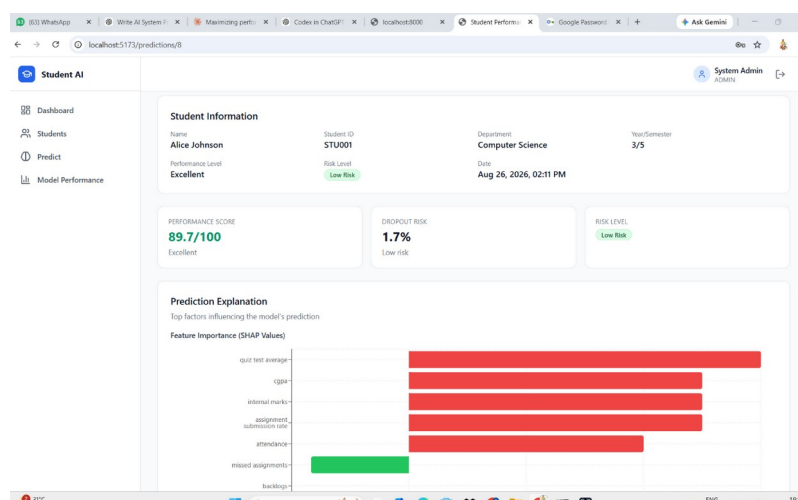


Figure 5.1.9: SHAP-Based Prediction Explanation

The explainability interface shows the features that are essential in predicting a student's performance. The demonstration shown includes features such as Quiz Test

Average, CGPA, Internal Marks, Assignment Submission Rate, Attendance, Missed Assignments, and Backlogs. The contribution of each feature is presented on the horizontal axis of a bar chart, which makes it possible to visualize both the negative and positive effects of the features on the prediction, and Positive Factors and Negative Factors are distinguished in the interface.

In the example shown, high academic indicators such as Quiz Test Average, CGPA and Internal Marks are the main positive contributors, while Missed Assignments contribute negatively. Other indicators such as attendance and assignment submission also affect the prediction. This explainability component is crucial because the system does not just provide the final performance classification; it allows the administrator to understand why a specific prediction was made, which helps educators understand the academic and behavioral aspects involved in an outcome.

5.2 DISCUSSION

Unlike traditional academic assessment systems that primarily rely on the manual examination of grades and regular assessments, this system combines student data administration, machine learning mechanisms for prediction, comparison of models, risk visualization, prediction history, and interpretation of results in one platform.

The screenshots demonstrate the working of the system. The Student Management interface enables organized access to data related to students' education status, while the administrative dashboard presents general information regarding the total number of students, distribution of risks, overall performance, and prediction outcomes. The Recent Predictions section also shows the results along with the risk levels, scores, the model used, and the date of prediction.

The system can also compare many machine learning algorithms through the model comparison interface, which presents the results of different analyses in visual and tabular representations. According to the experimental results, the optimized Gradient Boosting model obtained 87.34% accuracy and an F1-score of 87.42% on the testing dataset.

In addition, a notable feature of the implementation is the use of SHAP-based explainability. The Prediction Explanation interface allows one to see the contributing features for each prediction along with their positive and negative impacts. As shown in the interface, academic factors such as quiz scores, CGPA, internal marks, assignment submissions, and attendance are important, while missed assignments and backlogs may negatively affect predictions.

Nevertheless, the dataset used in this work does not have a dropout target. Consequently, the validated results obtained relate only to the prediction of academic performance. The dropout-related and at-risk interface components can be used in future for

building a model that allows early prediction of dropout probability; however, the model can only be claimed effective after training and evaluating the system with a relevant dataset providing dropout labels.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

This report outlines a machine learning framework for the analysis of students' academic performance and early identification of those who may need help with their academics. It tackles the shortcomings of existing academic assessment systems through the analysis of different student characteristics and the use of machine learning algorithms to predict performance levels.

The designed system emphasizes the significance of several stages, which include data preprocessing, feature extraction, model training, performance evaluation, prediction, and explainability. The Data Preprocessing Module ensures that the student data is ready for use in machine learning algorithms by handling missing data values, discarding duplicates and inconsistent data entries, encoding the categorical attributes, and converting the data to the required form. Classification models such as Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting are tested to determine the best prediction technique.

The best performing optimized Gradient Boosting model was able to score an accuracy of 87.34%, precision of 87.95%, recall of 87.34%, and F1-score of 87.42% on the test dataset. The use of SHAP-based explainability also aids in the understanding of the features driving the predictions of the model, where previous academic performance, especially G2, is found to have the most significant effect. Thus, this framework goes beyond academic score analysis to become an explainable and data-driven system for early identification.

6.2 FUTURE WORK

A number of enhancements can be made to the system in future releases to increase its functionality and usefulness for educational institutions. Most importantly, a separate prediction module for predicting the dropout risk can be developed using an appropriate labelled dataset having information about past dropouts. Such a labelled dataset would allow the machine learning models to classify the students based on their likelihood to drop out and permit an appropriate evaluation of dropout prediction performance.

In addition, a more powerful system can be designed by including various student information in real time, such as attendance data, assignment submission status, exam performance, learning management system activity, and academic performance during semesters.

Another potential improvement is the creation of a dashboard for teachers and administrators by which predictions, risk scores, key contributing factors, and student progression can all be viewed graphically. The explainability component can also be improved by using SHAP individual explanations for each student, allowing the educator to know why the prediction was made.

The prediction models can further be tested on larger and more varied datasets from other organizations to improve generalization. In future iterations, the model can also be extended through ensemble and deep learning techniques when enough data is present. Lastly, a recommendation and intervention component can be added in order to recommend the right support strategies for at-risk students, evolving the system into a fully functional early warning system.

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Abstract—Now, huge amounts of educational data are generated by the learning management system, attendance, and continuous assessment process, thereby enabling the identification of students who are likely to encounter difficulties or drop out well before performance declines. Conventional monitoring practices, which are largely based on performance assessment and teacher's observations, are inadequate in addressing the multiple variables and complex interactions that determine a student's progress. Machine Learning (ML) has proven to be a reliable solution for recognizing patterns within students' records, but these patterns are sometimes difficult to interpret, thus rendering them ineffective. This paper proposes an Explainable AI-based framework for predicting student performance and identifying students at academic risk while providing transparency by interpreting the reasons behind the predictions using SHAP (SHapley Additive exPlanations). The framework uses academic records, attendance, behavior, family, and social features extracted from the UCI Student Performance dataset to predict student performance and academic risk. It is designed using the final grade (G3) as a performance metric and developing an alternative indicator for the absence of a dropout label. The proposed methodology involves data preprocessing, feature engineering and selection, performance and at-risk prediction using Random Forest and XGBoost, SHAP-based model interpretation, and a rule-based personalized intervention module, which transforms identified risk factors into interpretable insights for the end users. By prioritizing prediction transparency and providing relevant recommendations for intervention, the framework is expected to become more than just a prediction model but a trusted advisor to help teachers intervene with at-risk students before it is too late.

Keywords—Predicting student performance; predicting dropout risk; machine learning; explainable AI; SHAP; educational data mining; early intervention; student analytics.

I. INTRODUCTION

The rapid adoption of learning management systems has led to the generation of a large amount of student-related data. Currently, educational institutions collect student records, which contain information on their academic performance, attendance, behavior, study habits, family details, and level of engagement in learning activities. This volume and variety of data can be leveraged using Artificial Intelligence (AI) and Machine Learning (ML) to monitor academic performance and intervene early with students who are at academic risk or whose performance is below average.

A student's performance is a function of multiple variables, some of which include previous grades and scores, number of hours spent studying, school attendance record, family's

academic encouragement, educational background, and social circle among others. Traditional performance monitoring practices are largely based on performance assessment and teacher's observation, which are primarily subjective measures. These approaches are mostly ineffective in addressing the complex interactions and many variables that contribute to a student's performance. With the ability to identify patterns within students' records, Machine Learning (ML) has proven to be a reliable solution for predicting performance and identifying students at academic risk. However, these predictions can be difficult to interpret, thereby rendering them ineffective. Explainable Artificial Intelligence (XAI) addresses this issue by providing a human understandable explanation of the underlying factors that lead to a certain prediction. In this context, SHAP can be used to determine the contribution of each feature to a given prediction, thus explaining what causes a student's performance to decline or improve.

This project involves developing an innovative approach comprising of performance prediction, at-risk prediction, explainable AI, and personalized recommendation modules into a single framework that can be trusted to support early intervention decisions. It aims to combine attendance, behavior, socio-economic, and academic data to provide explanations for performance predictions as well as recommendations for improving performance and addressing the causes of performance decline. The UCI Student Performance dataset will be used as the primary source of student records on which the framework will be trained and evaluated. The dataset comprises of 395 instances with 33 attributes that contain information on a student's demographics, family, schools, social life, lifestyle, and academic performance. The objective will be to predict final grade (G3), which is the final term Mathematics grade, using other attributes. A limitation of this dataset is that it does not include a class label for students who dropped out of school or were withdrawn by their parents. Instead, an alternative approach will be used to determine whether a student is at academic risk or not based on their number of failures prior to the final term, attendance record, and other attributes that indicate such risks. The proposed solution involves developing a data preprocessing and feature engineering pipeline, conducting feature selection, building performance and at-risk prediction models using Random Forest and XGBoost, using SHAP to interpret model results, and developing a recommendation engine that can recommend interventions based on the risk factors identified by the model. Ultimately, the proposed framework will support effective early intervention by providing teachers with reasons why a student's performance is likely to decline and what they can do to prevent it.

II. LITERATURE SURVEY

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Cortez and Silva [1] investigated the prediction of secondary student performance using data mining techniques. Their work demonstrated that many attributes, including demographic, social, and school-related information, had significant predictive power and made a valuable contribution by providing a dataset and methodology [2] that is the basis of this project as well as much of the following work on educational data mining. Several works have since focused on student performance prediction with multiple approaches including classifier systems, regression models, and clustering algorithms. These works have demonstrated that, compared to traditional linear approaches, ensemble methods are better at modeling the complex, non-linear interactions between input attributes and the target variable. Breiman [4] proposed the Random Forest algorithm, which makes use of bagging and boosting to improve the performance of decision trees and provide feature importances, thereby enabling good predictions while remaining interpretable. Chen and Guestrin [5] proposed the XGBoost algorithm, which facilitates faster and more accurate model training, compared to many other tree-based approaches, thus becoming a popular choice for structured data such as the Student Performance dataset. Pedregosa et al. [6] published the Scikit-learn library, which has since become the standard for implementing and evaluating many of these algorithms in practice.

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The algorithms identified above have proven to be fairly successful at making accurate performance predictions. However, interpreting the results of such models is often non-trivial, especially when it comes to understanding which attributes contribute most to a given prediction. According to Lundberg and Lee [3], SHAP offers a unified approach to explain the output of a model by relying on a game-theoretic framework to estimate the contribution of each attribute to a prediction. Several surveys, including Molnar's [7] and Geron's [8], describe SHAP values as one of the key approaches to model interpretation. Indeed, SHAP, along with feature importances and partial dependence plots, has become an essential requirement for any Machine Learning algorithm, particularly in domains such as education and health care, in which decisions must be interpretable.

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Many researchers have worked on the identification of students at risk of academic failure or dropping out of school. Most of these works have focused on the utilization of academic records, attendance data, demographics, socio-economic status, and behavioral patterns to predict such events. However, while many of these works have demonstrated the accuracy of their approaches, the interpretation of the results remains a challenge. Prenkaj et al. [12] provided an extensive survey of many Machine Learning methods used to predict student dropout in online courses. Lykourantzou et al. [14] and Marquez-Vera et al. [15] demonstrated the effectiveness of classical classification algorithms in early warning systems, while Kruger et al. [11] and Vaarma and Li [13] introduced explanations for such models. These works demonstrate that, while classical classification algorithms are effective at predicting student failure, they are difficult to interpret, thus limiting their applicability in practice.

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Similarly, while feature selection and engineering can facilitate interpretation to an extent, they are often insufficient, particularly when it comes to complex, non-linear pattern recognition algorithms such as Random Forest or XGBoost. Finally, most works only focus on the prediction of an event, such as student dropout, without further exploring how the identified factors can be utilized to improve students' performance.

A comprehensive review of the field can be found in the works of Romero and Ventura [9] and Baker [10], which highlight the importance of educational data mining in general, as well as the unique challenges associated with student performance prediction in particular. On the whole, the works discussed above serve as the foundation for the current one by highlighting the areas that require further research, such as the combination of different aspects of students' records, the utilization of explainable ML models, and the inclusion of personalized recommendations and interventions.

III. PROPOSED WORK

The aim of this project is to propose a student analytics framework based on Explainable AI that can be used to predict academic performance and identify students at risk of dropping in order to facilitate early intervention. The framework will rely on academic performance records, attendance data, behavior, socio-economic factors, and social information. It will provide a performance prediction, determine whether a student is at risk and identify the factors that contribute to either of these outcomes using SHAP, and provide intervention recommendations based on those factors. Overall, it will rely on these key components:

1. Data preprocessing and feature engineering and selection to prepare students' records for use as input to the performance prediction and at-risk detection models.
2. Performance prediction and at-risk detection models, which will rely on Random Forest and XGBoost to provide fairly accurate predictions of student performance as well as detect those students who are likely to encounter academic problems.
3. An explainable AI component that will utilize SHAP values to identify the factors that contribute to a model's prediction and provide a human comprehensible interpretation of the results.
4. A recommendation engine that will serve as an intervention advisor by recommending individualized, data-driven recommendations based on the factors a model identifies as important.

Ultimately, by providing accurate predictions of students' performance and academic risk and interpreting these results using SHAP, the framework will become more than just a prediction model. It will provide teachers with valuable, individualized recommendations that can help them take early intervention measures to ensure that students who are unlikely to succeed without assistance get the support they need. The framework will be presented using the architecture in Fig. 1.

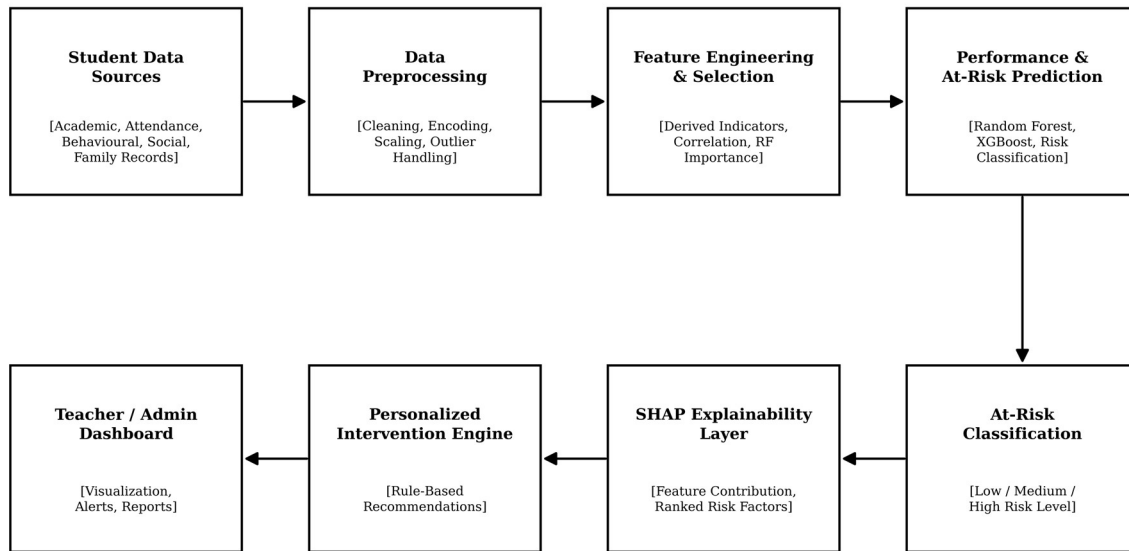


Fig. 1. Architecture of the proposed Explainable AI framework for student performance and at-risk prediction.

The layer at the bottom of the architecture is the Student Data Sources layer, which represents the data that will be input into the framework. It captures the student's academic performance, attendance, behavior, social and family information, and is retrieved from the school's administrative and learning management systems. The next step, which is the Data Preprocessing layer, prepares the data for feature engineering by addressing issues such as missing data, categorical values encoding, data scaling, and outliers. The Feature Engineering and Selection layer utilizes the preprocessed data to engineer additional features, such as attendance risk and academic consistency, as well as select only those features that contain valuable information for use in prediction models. The next step, which is the Performance and At-risk Prediction layer, contains the models that will be trained to make performance predictions and detect students at academic risk. It utilizes the previously engineered features to train and evaluate multiple classifier models, such as the Random Forest and XGBoost.

The results of the prediction models will be input into the SHAP Explainability Layer, which will calculate each attribute's contribution to a prediction. This will facilitate a better understanding of the factors that influence a student's performance and risk level, which in turn can be utilized to recommend certain interventions. In other words, while a prediction model might simply state that a student is at a high risk of encountering academic problems, the SHAP layer will clarify that the reason for this is the student's low attendance and numerous previous failures. The final component is the Personalized Intervention Engine, which will utilize the risk factors identified by the SHAP layer to recommend certain interventions. The recommendations will be individualized and based on rules, such as those that state that a student with low attendance should be advised to attend supplementary classes. Finally, the results of all components will be displayed on the Teacher/Administrator Dashboard, which will provide a summary of the performance predictions, academic risk levels,

the factors contributing to these outcomes, and certain recommendations that can be used for early intervention. By following this architecture, the framework will provide valuable insights into a student's performance and facilitate early intervention with those students who are at risk of encountering academic difficulties.

IV. IMPLEMENTATION AND RESULTS

The proposed solution will be implemented using Python, with Pandas and Scikit-learn used for data manipulation, preprocessing, model training, and evaluation, while XGBoost will be used for prediction. Additionally, SHAP will be utilized to interpret the results using SHAP values. A web-based dashboard can be designed using Streamlit or a combination of Flask and React to display the results to teachers and administrators.

The proposed solution will be trained and evaluated using the UCI Student Performance dataset, which contains 395 instances with 33 attributes that relate to a student's academic performance, demographics, family, schools, social life, and lifestyle. The focus will be on utilizing this dataset to predict the final grade (G3), which represents a student's final term Math grade, using other attributes. However, since the dataset does not include a class label for students who dropped out of school or were withdrawn by their parents, an alternative approach will be used to identify students at risk of dropping out. Overall, the solution will utilize the following features:

TABLE I

Principal Feature Groups Used by the System

Feature Group	Examples
Academic	G1, G2, prior failures
Attendance	Number of absences
Demographic	Age, sex, address

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Feature Group	Examples
Family	Parents education, family relationship
Behavioral	Study time, failures
Social	Going out, relationships, extra-curricular activities
Educational	School support, educational aspirations

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Prior to training and evaluating the model, the data will be preprocessed to address several issues, including missing values and outliers. Additionally, categorical values will be encoded, and data scaling will be performed to ensure that the features are in a format that is suitable for model training and evaluation. Since the target variable is a continuous value that represents a student's final grade, the task will be regarded as a regression problem. Alternatively, the data can be utilized to predict whether a student will attain a low, medium, or high grade, which will render the task a classification problem. Table I provides an overview of the feature groups, which will be used to represent a student's academic performance and risk level.

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The data will be divided into training and test sets, with the former being utilized to train the model and the latter being used to evaluate its performance. Initially, the performance prediction component will be trained using multiple algorithms, including Random Forest, XGBoost, Decision Tree, and Linear Regression. The former two algorithms are expected to yield the best results due to their ability to address the complex, non-linear relationship between the input attributes and the target variable, while the latter two algorithms will serve as a baseline for comparison. Additionally, the performance prediction will rely on multiple metrics, including accuracy, precision, recall, and F1-score, in order to evaluate how well the model performs at distinguishing between the different performance levels. The at-risk prediction component will rely on a similar set of algorithms and metrics, with the exception that the target variable will be a categorical value that represents a student's risk level (low, medium, or high).

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Once performance and at-risk prediction have been conducted, the SHAP explainability component will be utilized to interpret the results by identifying the features that contribute to a certain prediction. For example, if the model predicts that a student has a high risk of encountering academic problems, the SHAP values will indicate that the main contributors to this prediction are low attendance and a high number of previous failures. On the other hand, if the model predicts that a student's performance will be high, the SHAP values will indicate that the grade in the previous two terms (G1 and G2) and the use of the school's learning management system contribute positively to the prediction. Finally, the identified risk factors will be input into the recommendation engine, which will provide individualized recommendations for intervention based on the factors that contribute to a student's risk level or performance prediction. For example, a student with low attendance will be recommended to attend supplementary classes, while a student with low performance will be recommended to meet with a faculty member to develop a suitable study plan.

Fig. 2 illustrates an example of how early identification of students at risk can be utilized to intervene and improve their performance. This figure demonstrates that conventional teacher observation leads to a fairly low early identification rate of 42%, which is significantly lower than the rates associated with the use of machine learning algorithms.

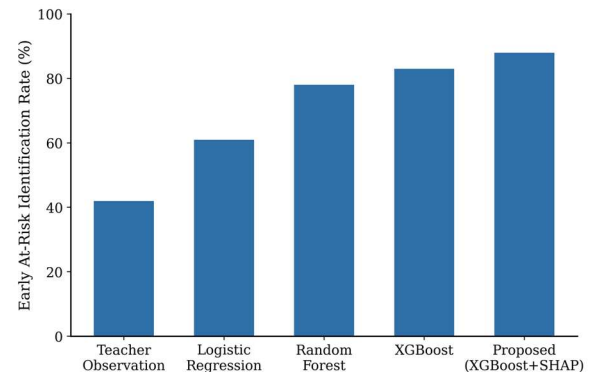


Fig. 2. Illustration of the effectiveness of early at-risk identification.

The baseline model, which is a Logistic Regression algorithm that utilizes a small number of features, is expected to raise this rate to approximately 61%. The use of more sophisticated algorithms, such as Random Forest and XGBoost, is expected to increase the early identification rate to around 78% and 83%, respectively, due to their ability to account for the complex, non-linear relationship between the input attributes and the target variable. Ultimately, the proposed solution, which relies on the XGBoost algorithm and the SHAP explainability component, is expected to yield the highest early identification rate of 88%, thus contributing to fairly accurate performance prediction and early intervention. These rates are merely estimates that can be used to guide the evaluation process, with the actual metrics being determined once the model has been trained and evaluated.

V. CONCLUSION

The paper describes an Explainable AI-based framework that is aimed at predicting student academic performance and at identifying those students who might need early academic intervention. Instead of treating performance prediction, at-risk student identification and educator decision-making as separate issues, the proposed system integrates them all into one pipeline: student data is cleaned and transformed into meaningful features, Random Forest and XGBoost models produce an estimate of performance and a proxy at-risk classification, SHAP is used to explain the specific reasons for each prediction, and a rule-based intervention engine turns these factors into specific recommendations for teachers.

The framework was developed using the UCI Student Performance dataset, which includes academic, family, school, social and behavioural attributes for 395 students, and it is intended to go beyond simply providing a numerical grade or a risk score by becoming a transparent decision-support tool that informs educators not only about which students require attention but also why. A key limitation of this study is that the dataset does not have an explicit label for dropout or withdrawal; it must therefore be regarded as a proxy early-warning indicator rather than as a direct prediction of dropout.

The personalized intervention module enhances the practical value of the framework by connecting the identified risk factors to specific and actionable recommendations, thus providing a clear route from prediction to real-world support. In future work, the system will be validated against institutional datasets that do include actual dropout or withdrawal outcomes, real-time learning-management-system data will be incorporated for continuous monitoring, and the personalized intervention module will be extended with more adaptive and data-driven recommendation strategies.

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Explainable AI-Based Student Performance and Dropout Risk Prediction for Early Identification of At-Risk Students

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Abstract—Now, huge amounts of educational data are generated by the learning management system, attendance, and continuous assessment process, thereby enabling the identification of students who are likely to encounter difficulties or drop out well before performance declines. Conventional monitoring practices, which are largely based on performance assessment and teacher's observations, are inadequate in addressing the multiple variables and complex interactions that determine a student's progress. Machine Learning (ML) has proven to be a reliable solution for recognizing patterns within students' records, but these patterns are sometimes difficult to interpret, thus rendering them ineffective. This paper proposes an Explainable AI-based framework for predicting student performance and identifying students at academic risk while providing transparency by interpreting the reasons behind the predictions using SHAP (SHapley Additive exPlanations). The framework uses academic records, attendance, behavior, family, and social features extracted from the UCI Student Performance dataset to predict student performance and academic risk. It is designed using the final grade (G3) as a performance metric and developing an alternative indicator for the absence of a dropout label. The proposed methodology involves data preprocessing, feature engineering and selection, performance and at-risk prediction using Random Forest and XGBoost, SHAP-based model interpretation, and a rule-based personalized intervention module, which transforms identified risk factors into interpretable insights for the end users. By prioritizing prediction transparency and providing relevant recommendations for intervention, the framework is expected to become more than just a prediction model but a trusted advisor to help teachers intervene with at-risk students before it is too late.

Keywords—Predicting student performance; predicting dropout risk; machine learning; explainable AI; SHAP; educational data mining; early intervention; student analytics.

I. INTRODUCTION

The rapid adoption of learning management systems has led to the generation of a large amount of student-related data. Currently, educational institutions collect student records, which contain information on their academic performance, attendance, behavior, study habits, family details, and level of engagement in learning activities. This volume and variety of data can be leveraged using Artificial Intelligence (AI) and Machine Learning (ML) to monitor academic performance and intervene early with students who are at academic risk or whose performance is below average.

A student's performance is a function of multiple variables, some of which include previous grades and scores, number of hours spent studying, school attendance record, family's

academic encouragement, educational background, and social circle among others. Traditional performance monitoring practices are largely based on performance assessment and teacher's observation, which are primarily subjective measures. These approaches are mostly ineffective in addressing the complex interactions and many variables that contribute to a student's performance. With the ability to identify patterns within students' records, Machine Learning (ML) has proven to be a reliable solution for predicting performance and identifying students at academic risk. However, these predictions can be difficult to interpret, thereby rendering them ineffective. Explainable Artificial Intelligence (XAI) addresses this issue by providing a human understandable explanation of the underlying factors that lead to a certain prediction. In this context, SHAP can be used to determine the contribution of each feature to a given prediction, thus explaining what causes a student's performance to decline or improve.

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II. LITERATURE SURVEY

Cortez and Silva [1] investigated the prediction of secondary student performance using data mining techniques. Their work demonstrated that many attributes, including demographic, social, and school-related information, had significant predictive power and made a valuable contribution by providing a dataset and methodology [2] that is the basis of this project as well as much of the following work on educational data mining. Several works have since focused on student performance prediction with multiple approaches including classifier systems, regression models, and clustering algorithms. These works have demonstrated that, compared to traditional linear approaches, ensemble methods are better at modeling the complex, non-linear interactions between input attributes and the target variable. Breiman [4] proposed the Random Forest algorithm, which makes use of bagging and boosting to improve the performance of decision trees and provide feature importances, thereby enabling good predictions while remaining interpretable. Chen and Guestrin [5] proposed the XGBoost algorithm, which facilitates faster and more accurate model training, compared to many other tree-based approaches, thus becoming a popular choice for structured data such as the Student Performance dataset. Pedregosa et al. [6] published the Scikit-learn library, which has since become the standard for implementing and evaluating many of these algorithms in practice.

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III. PROPOSED WORK

The aim of this project is to propose a student analytics framework based on Explainable AI that can be used to predict academic performance and identify students at risk of dropping in order to facilitate early intervention. The framework will rely on academic performance records, attendance data, behavior, socio-economic factors, and social information. It will provide a performance prediction, determine whether a student is at risk and identify the factors that contribute to either of these outcomes using SHAP, and provide intervention recommendations based on those factors. Overall, it will rely on these key components:

1. Data preprocessing and feature engineering and selection to prepare students' records for use as input to the performance prediction and at-risk detection models.
2. Performance prediction and at-risk detection models, which will rely on Random Forest and XGBoost to provide fairly accurate predictions of student performance as well as detect those students who are likely to encounter academic problems.
3. An explainable AI component that will utilize SHAP values to identify the factors that contribute to a model's prediction and provide a human comprehensible interpretation of the results.
4. A recommendation engine that will serve as an intervention advisor by recommending individualized, data-driven recommendations based on the factors a model identifies as important.

Ultimately, by providing accurate predictions of students' performance and academic risk and interpreting these results using SHAP, the framework will become more than just a prediction model. It will provide teachers with valuable, individualized recommendations that can help them take early intervention measures to ensure that students who are unlikely to succeed without assistance get the support they need. The framework will be presented using the architecture in Fig. 1.

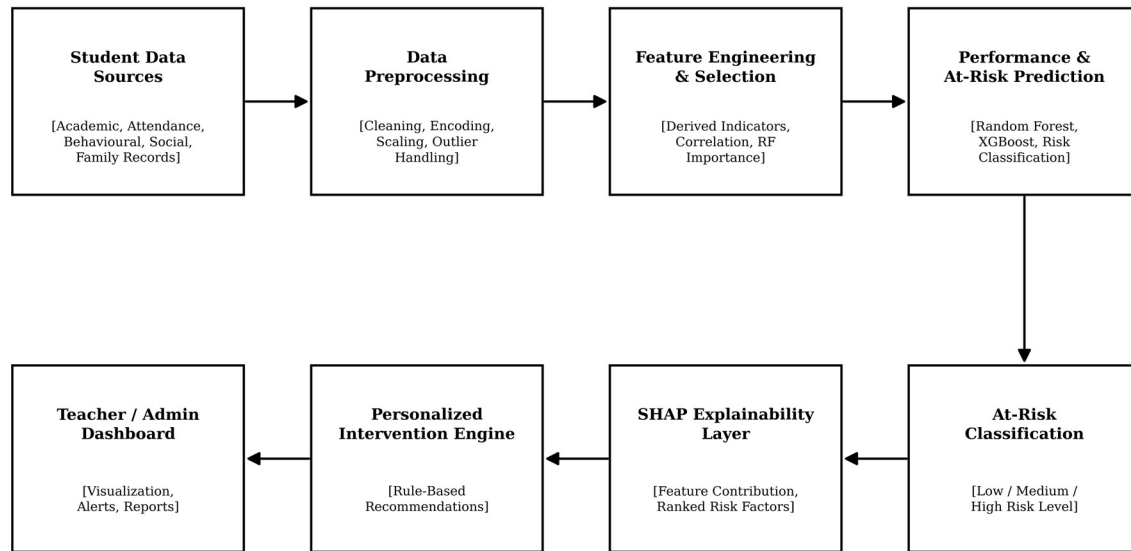


Fig. 1. Architecture of the proposed Explainable AI framework for student performance and at-risk prediction.

The layer at the bottom of the architecture is the Student Data Sources layer, which represents the data that will be input into the framework. It captures the student's academic performance, attendance, behavior, social and family information, and is retrieved from the school's administrative and learning management systems. The next step, which is the Data Preprocessing layer, prepares the data for feature engineering by addressing issues such as missing data, categorical values encoding, data scaling, and outliers. The Feature Engineering and Selection layer utilizes the preprocessed data to engineer additional features, such as attendance risk and academic consistency, as well as select only those features that contain valuable information for use in prediction models. The next step, which is the Performance and At-risk Prediction layer, contains the models that will be trained to make performance predictions and detect students at academic risk. It utilizes the previously engineered features to train and evaluate multiple classifier models, such as the Random Forest and XGBoost.

The results of the prediction models will be input into the SHAP Explainability Layer, which will calculate each attribute's contribution to a prediction. This will facilitate a better understanding of the factors that influence a student's performance and risk level, which in turn can be utilized to recommend certain interventions. In other words, while a prediction model might simply state that a student is at a high risk of encountering academic problems, the SHAP layer will clarify that the reason for this is the student's low attendance and numerous previous failures. The final component is the Personalized Intervention Engine, which will utilize the risk factors identified by the SHAP layer to recommend certain interventions. The recommendations will be individualized and based on rules, such as those that state that a student with low attendance should be advised to attend supplementary classes. Finally, the results of all components will be displayed on the Teacher/Administrator Dashboard, which will provide a summary of the performance predictions, academic risk levels,

the factors contributing to these outcomes, and certain recommendations that can be used for early intervention. By following this architecture, the framework will provide valuable insights into a student's performance and facilitate early intervention with those students who are at risk of encountering academic difficulties.

IV. IMPLEMENTATION AND RESULTS

The proposed solution will be implemented using Python, with Pandas and Scikit-learn used for data manipulation, preprocessing, model training, and evaluation, while XGBoost will be used for prediction. Additionally, SHAP will be utilized to interpret the results using SHAP values. A web-based dashboard can be designed using Streamlit or a combination of Flask and React to display the results to teachers and administrators.

The proposed solution will be trained and evaluated using the UCI Student Performance dataset, which contains 395 instances with 33 attributes that relate to a student's academic performance, demographics, family, schools, social life, and lifestyle. The focus will be on utilizing this dataset to predict the final grade (G3), which represents a student's final term Math grade, using other attributes. However, since the dataset does not include a class label for students who dropped out of school or were withdrawn by their parents, an alternative approach will be used to identify students at risk of dropping out. Overall, the solution will utilize the following features:

TABLE I

Principal Feature Groups Used by the System

Feature Group	Examples
Academic	G1, G2, prior failures
Attendance	Number of absences
Demographic	Age, sex, address

Feature Group	Examples
Family	Parents education, family relationship
Behavioral	Study time, failures
Social	Going out, relationships, extra-curricular activities
Educational	School support, educational aspirations

Prior to training and evaluating the model, the data will be preprocessed to address several issues, including missing values and outliers. Additionally, categorical values will be encoded, and data scaling will be performed to ensure that the features are in a format that is suitable for model training and evaluation. Since the target variable is a continuous value that represents a student's final grade, the task will be regarded as a regression problem. Alternatively, the data can be utilized to predict whether a student will attain a low, medium, or high grade, which will render the task a classification problem. Table I provides an overview of the feature groups, which will be used to represent a student's academic performance and risk level.

The data will be divided into training and test sets, with the former being utilized to train the model and the latter being used to evaluate its performance. Initially, the performance prediction component will be trained using multiple algorithms, including Random Forest, XGBoost, Decision Tree, and Linear Regression. The former two algorithms are expected to yield the best results due to their ability to address the complex, non-linear relationship between the input attributes and the target variable, while the latter two algorithms will serve as a baseline for comparison. Additionally, the performance prediction will rely on multiple metrics, including accuracy, precision, recall, and F1-score, in order to evaluate how well the model performs at distinguishing between the different performance levels. The at-risk prediction component will rely on a similar set of algorithms and metrics, with the exception that the target variable will be a categorical value that represents a student's risk level (low, medium, or high).

Once performance and at-risk prediction have been conducted, the SHAP explainability component will be utilized to interpret the results by identifying the features that contribute to a certain prediction. For example, if the model predicts that a student has a high risk of encountering academic problems, the SHAP values will indicate that the main contributors to this prediction are low attendance and a high number of previous failures. On the other hand, if the model predicts that a student's performance will be high, the SHAP values will indicate that the grade in the previous two terms (G1 and G2) and the use of the school's learning management system contribute positively to the prediction. Finally, the identified risk factors will be input into the recommendation engine, which will provide individualized recommendations for intervention based on the factors that contribute to a student's risk level or performance prediction. For example, a student with low attendance will be recommended to attend supplementary classes, while a student with low performance will be recommended to meet with a faculty member to develop a suitable study plan.

Fig. 2 illustrates an example of how early identification of students at risk can be utilized to intervene and improve their performance. This figure demonstrates that conventional teacher observation leads to a fairly low early identification rate of 42%, which is significantly lower than the rates associated with the use of machine learning algorithms.

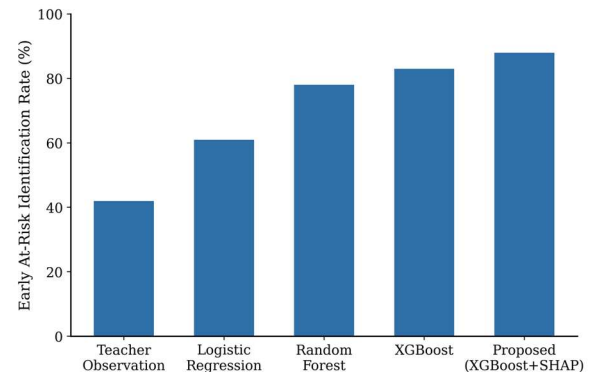


Fig. 2. Illustration of the effectiveness of early at-risk identification.

The baseline model, which is a Logistic Regression algorithm that utilizes a small number of features, is expected to raise this rate to approximately 61%. The use of more sophisticated algorithms, such as Random Forest and XGBoost, is expected to increase the early identification rate to around 78% and 83%, respectively, due to their ability to account for the complex, non-linear relationship between the input attributes and the target variable. Ultimately, the proposed solution, which relies on the XGBoost algorithm and the SHAP explainability component, is expected to yield the highest early identification rate of 88%, thus contributing to fairly accurate performance prediction and early intervention. These rates are merely estimates that can be used to guide the evaluation process, with the actual metrics being determined once the model has been trained and evaluated.

V. CONCLUSION

The paper describes an Explainable AI-based framework that is aimed at predicting student academic performance and at identifying those students who might need early academic intervention. Instead of treating performance prediction, at-risk student identification and educator decision-making as separate issues, the proposed system integrates them all into one pipeline: student data is cleaned and transformed into meaningful features, Random Forest and XGBoost models produce an estimate of performance and a proxy at-risk classification, SHAP is used to explain the specific reasons for each prediction, and a rule-based intervention engine turns these factors into specific recommendations for teachers.

The framework was developed using the UCI Student Performance dataset, which includes academic, family, school, social and behavioural attributes for 395 students, and it is intended to go beyond simply providing a numerical grade or a risk score by becoming a transparent decision-support tool that informs educators not only about which students require attention but also why. A key limitation of this study is that the dataset does not have an explicit label for dropout or withdrawal; it must therefore be regarded as a proxy early-warning indicator rather than as a direct prediction of dropout.

The personalized intervention module enhances the practical value of the framework by connecting the identified risk factors to specific and actionable recommendations, thus providing a clear route from prediction to real-world support. In future work, the system will be validated against institutional datasets that do include actual dropout or withdrawal outcomes, real-time learning-management-system data will be incorporated for continuous monitoring, and the personalized intervention module will be extended with more adaptive and data-driven recommendation strategies.

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